

DHCP

At the end of this episode, I will be able to:

1. Describe the Dynamic Host Configuration Protocol (DHCP) and discuss key aspects of its configuration
2. Describe the IPv6 configuration option of stateless address autoconfiguration (SLAAC)

Learner Objective: *Learn about the DHCP protocol and key aspects of its configuration*

Description: In this episode, you will learn about the role and functions of Dynamic Host Configuration Protocol (DHCP) in modern networks. You will also learn about stateless address autoconfiguration (SLAAC) capabilities in IPv6 networks.

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- The Dynamic Host Configuration Protocol (DHCP) is a network protocol that automates the assignment of IP addresses and other configuration parameters to devices on a network. DHCP allows devices to dynamically obtain network settings, such as IP addresses, subnet masks, and default gateways, streamlining the process of connecting to and communicating within a network.
 - A DHCP reservation is a configuration in a Dynamic Host Configuration Protocol (DHCP) server that associates a specific IP address with a particular device based on its Media Access Control (MAC) address, ensuring the consistent assignment of the reserved IP address to that device on the network.
 - A DHCP scope is a range of IP addresses configured on a Dynamic Host Configuration Protocol (DHCP) server, specifying the pool of addresses available for automatic assignment to devices on a network.
 - DHCP lease time is the duration for which an IP address is temporarily assigned to a device by a Dynamic Host Configuration Protocol (DHCP) server, specifying how long the device can use that address before it must renew or request a new lease.
 - DHCP options are additional parameters and configuration settings that a Dynamic Host Configuration Protocol (DHCP) server can provide to DHCP clients along with the basic network information, allowing for customization of client behavior and network configurations.
 - A DHCP relay is a network device or software feature that forwards Dynamic Host Configuration Protocol (DHCP) messages between DHCP clients and servers across different subnets, allowing clients in remote subnets to obtain IP addresses and network configuration information from DHCP servers.
 - DHCP exclusions are specific ranges of IP addresses set by a Dynamic Host Configuration Protocol (DHCP) server to be excluded from automatic assignment, ensuring that these addresses are not allocated to devices on the network to avoid conflicts or reserved for special purposes.
 - IPv6 Stateless Address Autoconfiguration (SLAAC) is a method that allows devices on a network to automatically configure their IPv6 addresses without the need for a centralized DHCP server by using information in router advertisements to generate unique addresses.
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Additional Resources:

Dynamic Host Configuration Protocol

https://en.wikipedia.org/wiki/Dynamic_Host_Configuration_Protocol